PRANALI RAJENDRA PATIL

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ABOUT

Computer Engineering student with a strong foundation in C++, Data Structures & Algorithms, and Software Development. Demonstrates a keen interest in solving complex problems through Machine Learning and computational logic. Proven ability to address hardware–software integration challenges, evidenced by securing All India Rank 3 in SAE e-BAJA and multiple hackathon achievements. Experienced in developing scalable web applications and data-driven solutions, with a strong aspiration to contribute effectively to high-performance software engineering teams

EDUCATION

Pimpri Chinchwad College of Engineering -Pune India
B.Tech in Computer Engineering | CGPA: 9.05| 2023-2027

HSC (12th Grade) | Score: 90%

SSC (10th Grade) | Score: 100%

TECHNICAL SKILLS

Languages: C++, C, Python, HTML, CSS

ML & AI: Machine Learning, Neural Networks, Deep Learning, Generative AI, XAI (SHAP, Grad-CAM++), LLMs, RAG

Embedded: ESP32, RTOS, CAN Bus, Arduino

Databases & Data: MySQL, MongoDB, SQL Query Optimization, Data Modeling, ETL Pipelines, Data **CleaningCloud & Tools:** Git, Firebase, Power BI, Jupyter Notebook, AWS cloud

TECHNICAL PROJECTS

PCOSMART – AI Healthcare Platform | Python | MERN | XAI |LLMs

Built an AI-driven PCOS prediction system integrating Explainable AI (XAI) for clinical transparency and LLM-powered personalized health recommendations.
Won 1st Place at SheSolve Hackathon; full-stack deployment with React + Node.js + Python ML backend.

Rice Leaf Disease Detection | Swin Transformers | ResNet50 | DCGAN | SHAP | Grad-CAM++

Achieved 95.33% accuracy detecting 8 rice diseases; resolved dataset imbalance via DCGAN-based synthetic image generation.
Applied Grad-CAM++ and SHAP for research-grade Explainable AI — validated model decision regions against agronomist labels.

Neural Network-Based EV Range Prediction | TensorFlow | Keras |CAN Bus

Modeled non-linear battery behavior using live CAN Bus telemetry; deep learning approach outperformed traditional regression on real-time range estimation.

Hospital Resource Optimization | DES | Machine Learning

Hybrid Discrete Event Simulation + ML framework optimized patient flow, reducing average waiting time by 20% over FCFS baseline with $R^2 = 0.86$.

NyayaAI – Constitutional Legal RAG System | LangChain | ChromaDB | Gemini Embeddings

Engineered end-to-end RAG pipeline: automated ingestion → chunking → vector indexing → citation-based retrieval for constitutional documents.
Enabled structured legal reasoning with source citations, reducing hallucination risk in LLM legal responses.

Canteen Optima – DAA Algorithm Visualizer | React.js | Node.js

Interactive visualizer for Dijkstra, Knapsack, TSP, Graph Coloring, Quick/Merge Sort, and Binary Search with animated step-by-step execution and playback controls.

Safe Route Navigator – Crime-Rate Path Finder | Data-Driven Navigation (Copyrighted)

Developed a risk-scored navigation algorithm using crime datasets and path optimization to identify safest travel routes.

Excel Analytics Platform | MERN Stack | ETL

Full-stack analytics platform with file ingestion, automated data transformation, interactive visualization pipelines, and historical data tracking.

KEY ACHIEVEMENTS

1st Place – SheSolve Hackathon 2026(Anantya)
AIR 3 – SAE e-BAJA (National)
5★ C++ & Problem Solving – HackerRank
Best Research Paper Award –ABHINAV YPC
85+ LeetCode | CodeChef 1363 Rating

EXPERIENCE

Data Acquisition Engineer
Team Red Baron – SAE e-BAJA | 2023–2025 | Pune

Designed a real-time EV data logging system using ESP32, CAN Bus, GPS, and GSM modules.
Implemented RTOS for concurrent multi-sensor data processing; monitored SoC, current, temperature, and vehicle telemetry.
Contributed core telemetry engineering to the team's All India Rank 3 finish — without any OEM DAQ hardware

Webmaster
ACM-W Student Chapter & R&D Club| 2024–2026 | Pune
Managed website development, deployment, and infrastructure. Built and maintained technical web systems